

- 7 (a) Angular velocity is the rate of change of angular displacement of a radius joining the body to the axis of rotation. **B1**

$$\omega = \frac{2\pi}{T}$$

(b)(i)
$$= \frac{2\pi}{(44.2 \times 365 \times 24 \times 3600)}$$
$$= 4.5 \times 10^{-9} \text{ rad s}^{-1}$$

C1

A1

- (ii) magnitude of the centripetal force about P is the same M1

$$M_1 x \omega^2 = M_2 (d - x) \omega^2$$

$$\frac{M_1}{M_2} = \frac{(d - x)}{x}$$

A1

- (iii) $x = 0.4d$ C1

Consider S_2 , gravitational force provides centripetal force

$$\frac{GM_1 M_2}{d^2} = M_2 (d - x) \omega^2$$

C1

$$GM_1 = d^2 (d - x) \omega^2$$

$$(6.67 \times 10^{-11}) M_1 = (1.0 - 0.4) (1.8 \times 10^{12})^3 (4.5 \times 10^{-9})^2$$

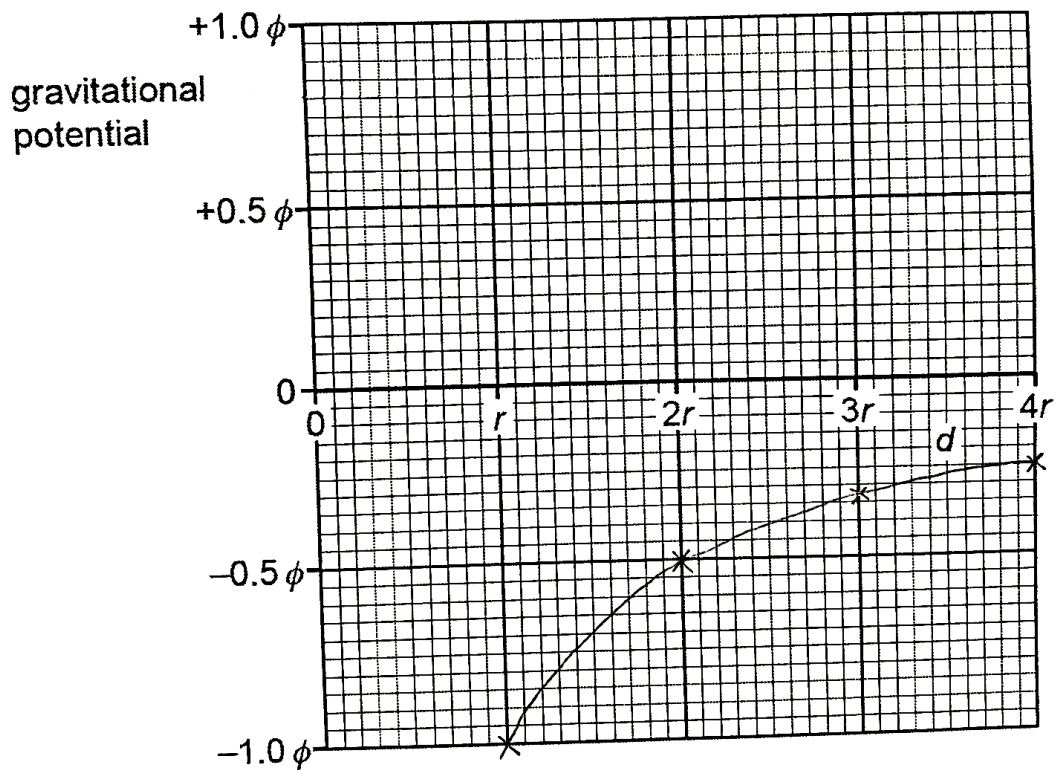
C1

$$M_1 = 1.1 \times 10^{30} \text{ kg}$$

A1

- (c)(i) The work done per unit mass in bringing a small test mass from infinity to that point. B1
B1

- (ii) curve from r to $4r$, with gradient of decreasing magnitude and line showing potential is negative throughout B1
 line passing through $(2r, -0.5\phi)$ and $(4r, -0.25\phi)$ B1



Section B

8 (a)(i) A photon is a discrete packet (quantum) of energy of electromagnetic radiation